Measurement-Free Error Correction via Multi-Qubit Syndrome Gate Encoding

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Abstract

Logical qubits can be protected from decoherence by performing quantum error correction. Naturally, the QEC algorithm requires feedback process to specific hardware platform which takes more time and cause extra error. In our sheme, we intend to develop a measurement-free error correction scheme based on surface code. With individually addressable, optically trapped, dual-species Rydberg atoms, we perform the multiqubit parity gates to map the message from data qubits to ancilla qubit and design a error-correction process without measurement feedback.

Keywords. measurement-free error correction · dual species · parity-gate · multi-qubit gate · etc

AMS subject classification. 65M70, 65M12

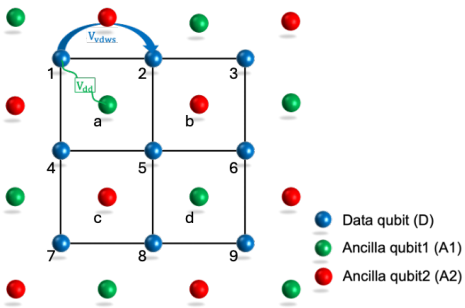
1 Motivation

(i) In order to eliminate the time required for quantum error correction process, we intend to develop a parity-controlled multiqubit gate to perform one-step syndrome mapping from data qubits to ancilla qubit.

(ii) After the syndrome mapping, there is a change in the ancilla qubit state if there is an error in data qubits. We will apply another pulse sequence to manipulate a C_2NOT gate to fix the error qubit. Thus, we do not need to perform measurement on the ancilla qubit, and the error correction process can be completed in one step.

2 Model(Overview)

The figure shown below is the measurement-free error correction model in toric code space. There are totally three species of atoms with green and red dots represent ancilla qubits and the blue dots represents the data qubits.



In the model shown above, we intend to perform an effective C_4NOT parity gate. If an Z error occurs in the data qubit, the neighboring red dots will flip. If an X error occurs in the data qubit the neighboring green dots will flip. Then, we intend to perform a C_2NOT gate where the ancilla qubits acts as control qubits, and the errored data qubit is the target qubit. The C_2NOT gate will correct the errored qubit without measuring process.

2.1 Interactions between atoms

In this multi-qubit platform, we intend a strong dipole-dipole interaction between different atom species and weak van der Waals interaction between atoms of the same species. ($V_{\text{dd}} \gg V_{\text{vdws}}$), where interactions between atoms of the same species are negligible. It largely reduces the error caused by undesirable interactions between data qubits and crosstalk between data qubits and ancilla qubits.

2.2 Effective parity check and error-correction

The effective syndrome mapping C_4NOT gate can be implemented through H-gate and CNOT$_4$ gate with the ancilla qubit in the center of the plaquette acts as the control qubit and the data qubits are acting as target qubits.

In our assumption, if there is no error, the data qubits and ancilla qubits should keep invariant after applying the pulse sequence.

If there is an error happens in the middle data qubit 5, different error types(X-error or Z-error) will influence different neighboring ancilla qubits. For example, if an Z-error happens in the data qubit 5, the neighboring ancilla qubits b and c will flip their states. Then, the error-correction process will perform a C_2NOT gate on the errored data qubit 5 with the ancilla qubits b and c as control qubits. If the ancilla qubits are in flipped state, the target data qubit can be corrected to the original state. If we assume that the initial state of the ancilla qubits are $|0\rangle$, the target data qubit will be corrected if the ancilla qubits are in state $|1\rangle$:

$$
\begin{aligned}
|0_A 0_A 0_T\rangle &\rightarrow |0_A 0_A 0_T\rangle \\
|0_A 0_A 1_T\rangle &\rightarrow |0_A 0_A 1_T\rangle \\
|0_A 1_A 0_T\rangle &\rightarrow |0_A 1_A 0_T\rangle \\
|0_A 1_A 1_T\rangle &\rightarrow |0_A 1_A 1_T\rangle \\
|1_A 0_A 0_T\rangle &\rightarrow |1_A 0_A 0_T\rangle \\
|1_A 0_A 1_T\rangle &\rightarrow |1_A 0_A 1_T\rangle \\
|1_A 1_A 0_T\rangle &\rightarrow |1_A 1_A 1_T\rangle \\
|1_A 1_A 1_T\rangle &\rightarrow |1_A 1_A 0_T\rangle
\end{aligned}
$$

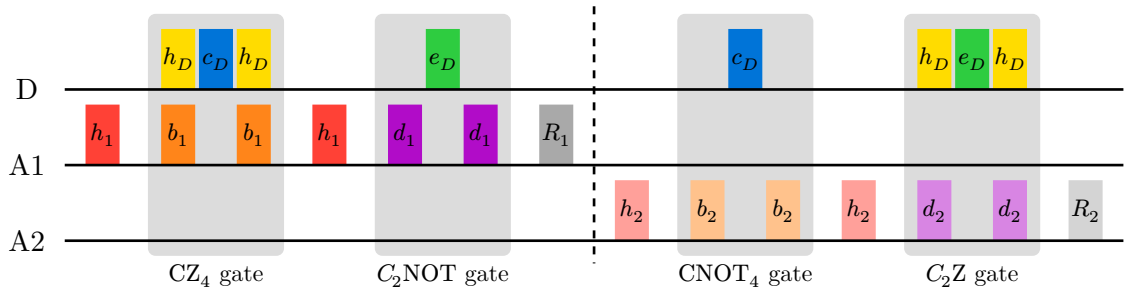
2.3 Why using different atom species?

Based on the multi-qubit CNOT$_k$ gate implementation[1], the fidelity is limited by the undesirable interaction between data qubits. Thus, choosing multi-species Rydberg atoms can overcoming these limits by using strong heteronuclear dipole-dipole interactions via Förster resonances for control and target atoms, while the target atoms are coupled by a weaker van der Waals interaction.

Also, the multi-species Rydberg array can avoid the crosstalk between data qubits and ancilla qubits.

2.4 Pulse sequence

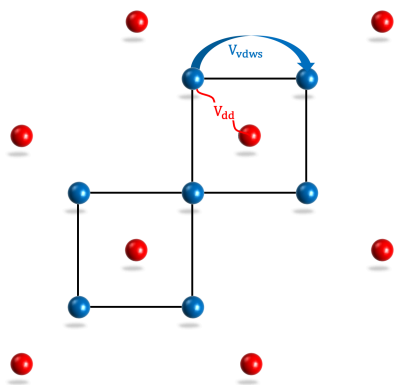
The pulse sequence is shown below:



Effect of different pulses

- Pulse $h_D/h_1/h_2$: act on D/A1/A2, Hadamard gate
- Pulse b_1/b_2: act on A1/A2, state transition $|1\rangle \leftrightarrow |r\rangle$
- Pulse c_D: act on D, if any neighboring A1 or A2 qubit in state $|r\rangle$, state transition $|0\rangle \leftrightarrow |1\rangle$, otherwise, do nothing. Note: it is a composite pulse, also known as the EIT pulse.
- Pulse d_1/d_2: act on A1/A2, state transition $|0\rangle \leftrightarrow |r\rangle$
- Pulse e_D: act on D, if any neighboring A1 or A2 qubit in state $|r\rangle$, do nothing, otherwise, state transition $|0\rangle \leftrightarrow |1\rangle$
- Pulse R_1/R_2: act on A1/A2, state transition $|?\rangle \to |0\rangle$ (reset)

In the first round of parity check process, the minimum unit is shown below:



2.5 Dual species architecture

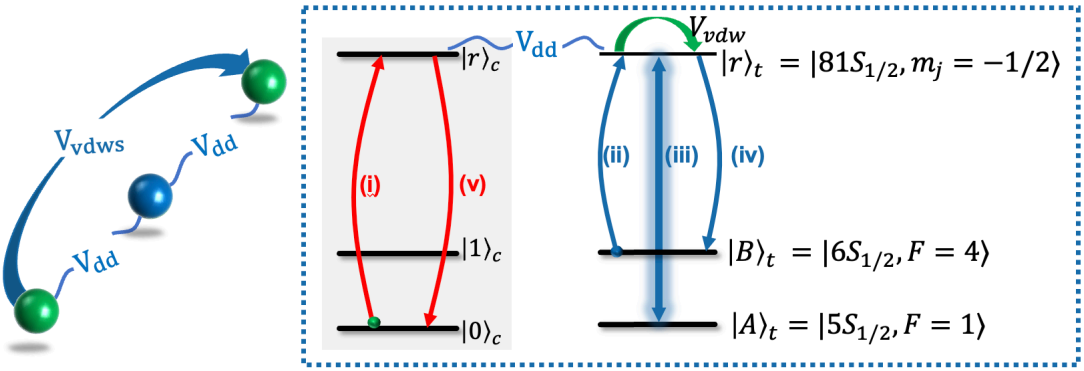
The interaction hamiltonian between control qubit and target qubits is given by:

$$h_{\text{ct}} = C_3(1 - 3\cos^2\theta_{\text{CT}})|r\rangle\langle r| \otimes |R\rangle\langle R|$$

where $C_3 = 2\pi \times 10\ \text{GHz} \cdot \mu m^3$, for target atoms in Rydberg state $|R\rangle$, $C_6 = 2\pi \times 2036\text{GHZ} \cdot \mu m^6$

2.6 Error-correcting C_2NOT gate construction

In this section, we intend to design a multi-qubit C_2NOT gate to correct the error qubit. The two ancilla qubits acts as a control qubit, and the errored data qubit is the target qubit. The model is shown below:



In this C_2NOT gate model, we assume a unbalanced interaction strength similar to the proposed CNOT$_4$ gate. The ancilla qubits(green dots) are control qubits and the errored data qubit(blue dot) is the target qubit. The interaction strength satisfies $V_{\text{dd}} \gg V_{\text{vdws}}$, where interactions between atoms of the same species are negligible. The pulse sequence is also shown from (i) to (v). All the pulses can be implemented with square pulse. The gate operation can be understood as follows:

(i) The control qubits are initially in the ground state $|0\rangle_c$. The first π-pulse (i) can couple it to the Rydberg state $|c\rangle_c$ and the target qubit is unchanged due to the strong dipole-dipole interactions.

(ii) The control qubits are initially in state $|1\rangle_c$. The first π-pulse (i) cannot couple the control qubits. Thus, the target qubit can be switched between state $|0\rangle_t$ and $|1\rangle_t$ through pulse sequence (ii) to (iv).

3 Literature Review

(i) Construction of multi-qubit CNOT$_k$ gate through EIT:[2], through optimization of the pulse[3], through adiabatic pulse design[1], [4] and experimental realization:[5], [6]; Parity gate construction through specially designed interaction strength[7].

(ii) Measurement-free fault-tolerant quantum error correction:[8]

(iii) Mid-circuit correction of correlated phase errors using an array of spectator qubits[9]

Bibliography

[1] A. M. Farouk, I. I. Beterov, P. Xu, S. Bergamini, and I. I. Ryabtsev, "Parallel Implementation of CNOTN and C2NOT2 Gates via Homonuclear and Heteronuclear Förster Interactions of Rydberg Atoms," *Photonics*, vol. 10, no. 11, 2023, doi: 10.3390/photonics10111280.

[2] J. M. Auger, S. Bergamini, and D. E. Browne, "Blueprint for fault-tolerant quantum computation with Rydberg atoms," *Phys. Rev. A*, vol. 96, no. 5, p. 52320, Nov. 2017, doi: 10.1103/PhysRevA.96.052320.

[3] J. Kazemi, M. Schuler, C. Ertler, and W. Lechner, "Multi-Qubit Parity Gates for Rydberg Atoms in Various Configurations." [Online]. Available: https://arxiv.org/abs/2506.09686

[4] C. Dlaska, K. Ender, G. B. Mbeng, A. Kruckenhauser, W. Lechner, and R. van Bijnen, "Quantum Optimization via Four-Body Rydberg Gates," *Phys. Rev. Lett.*, vol. 128, no. 12, p. 120503, Mar. 2022, doi: 10.1103/PhysRevLett.128.120503.

[5] C.-Y. Guo, L.-L. Yan, S. Zhang, S.-L. Su, and W. Li, "Optimized geometric quantum computation with a mesoscopic ensemble of Rydberg atoms," *Phys. Rev. A*, vol. 102, no. 4, p. 42607, Oct. 2020, doi: 10.1103/PhysRevA.102.042607.

[6] K. McDonnell, L. F. Keary, and J. D. Pritchard, "Demonstration of a Quantum Gate Using Electromagnetically Induced Transparency," *Phys. Rev. Lett.*, vol. 129, no. 20, p. 200501, Nov. 2022, doi: 10.1103/PhysRevLett.129.200501.

[7] F. Q. Guo, S. L. Su, W. Li, and X. Q. Shao, "Parity-controlled gate in a two-dimensional neutral-atom array." [Online]. Available: https://arxiv.org/abs/2405.19564

[8] S. Heußen, D. F. Locher, and M. Müller, "Measurement-Free Fault-Tolerant Quantum Error Correction in Near-Term Devices," *PRX Quantum*, vol. 5, no. 1, Feb. 2024, doi: 10.1103/prxquantum.5.010333.

[9] K. Singh, C. E. Bradley, S. Anand, V. Ramesh, R. White, and H. Bernien, "Mid-circuit correction of correlated phase errors using an array of spectator qubits," *Science*, vol. 380, no. 6651, pp. 1265–1269, 2023, doi: 10.1126/science.ade5337.